PLA

Polylactic Acid

195 products from 40 makers. A material type on Bambu Lab's H2C list, though not every brand of it is.

Tensile modulus (XY)	Tensile strength (XY)	Tensile strength (Z)	Notched Charpy impact	HDT at 0.45 MPa	Price
2.53 GPa	**45.5** MPa	**30.6** MPa	**6** kJ/m²	**55** °C	**33.7** CAD/kg
0.5 – 10 (log)	0 – 150	0 – 60	1 – 50 (log)	50 – 300	0 – 300
55 products middle half 1.93–2.87	71 products middle half 40–53	39 products middle half 22.9–34	14 products middle half 3.97–7.9	89 products middle half 53.8–59	26 products middle half 28–40

Well documented 281 documents from 41 publishers, 240 of them data sheets. Each figure is the median of the products that publish it with a stated orientation or test load, and beside it the middle half of them. Products with an added filler or additive, and for notched impact those sold as toughened, are shown but kept out of the medians.

PLA is polylactic acid. Makers sell it for prototypes and display models, and Prusa's Knowledge Base rates it easy to print. It is amorphous as printed, so it softens at its glass transition: its HDT at 0.45 MPa is 55 °C, and the heavier load gives the same. On the H2C it needs no heated chamber and, except a few filled products, no hardened nozzle. Makers' statements on acids and alkalis mostly say it does not resist them.

On the H2C

Nozzle	**195–230 °C** most products that give one
Bed	**50–60 °C** most products that give one
Heated chamber	**none needed** 160 of 195 products
Drying	**40–65 °C, 3–12 h** most advise it only for wet filament
Nozzle type	**standard for most** 6 of 195 products' sheets ask for one; 34 say a standard nozzle is enough; Bambu Lab's guide says any for 140 more; the rest are silent

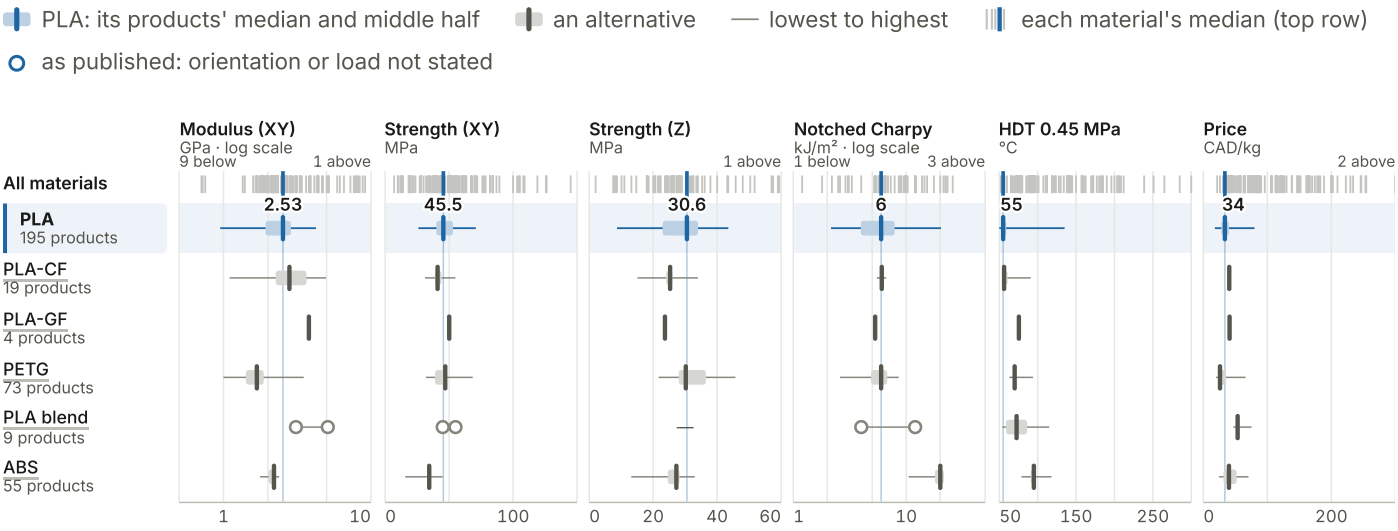
No product's setting is beyond the H2C's limits: nozzle 350 °C, bed 120 °C, heated chamber 65 °C.

Among the materials

Against PLA-CF, PLA is 21 % stronger across the layers, 17 % less expensive and 11 % stronger flat. Against PETG, PLA is 50 % stiffer and 29 % more expensive, and its HDT at 0.45 MPa is 15 °C lower. Among the materials, PLA has a low HDT at 0.45 MPa and price: both are in the lowest quarter of the materials that publish them. Its tensile modulus (XY), tensile strength (XY), tensile strength (Z), elongation at break (XY), notched Charpy impact and density sit in the middle half.

PLA against its alternatives

How does it compare with what an engineer would buy instead?



Each row is a material, each column a property on the scale it has on every page: the bar is the middle half of a material's products, the tick their median, the hairline their lowest to highest. The ticks above the rows are every material's median; those beyond the scale are counted at its ends, and a chevron marks a value beyond it.

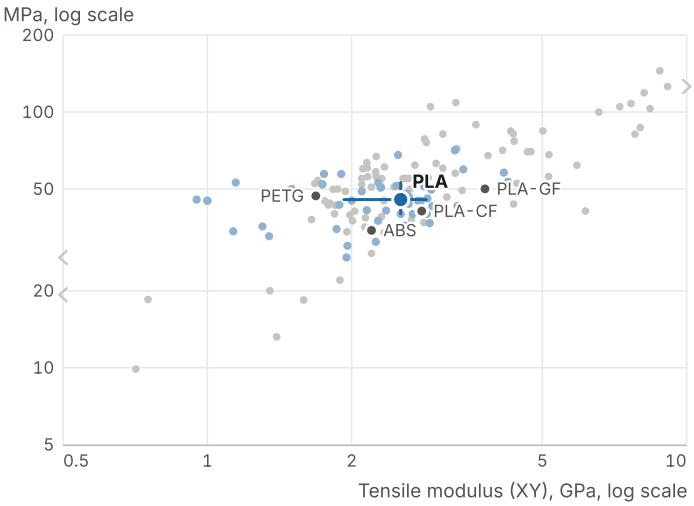
Where PLA sits among the 126 rigid materials on file

Where is it in the landscape: stiffness against strength, heat resistance against price?

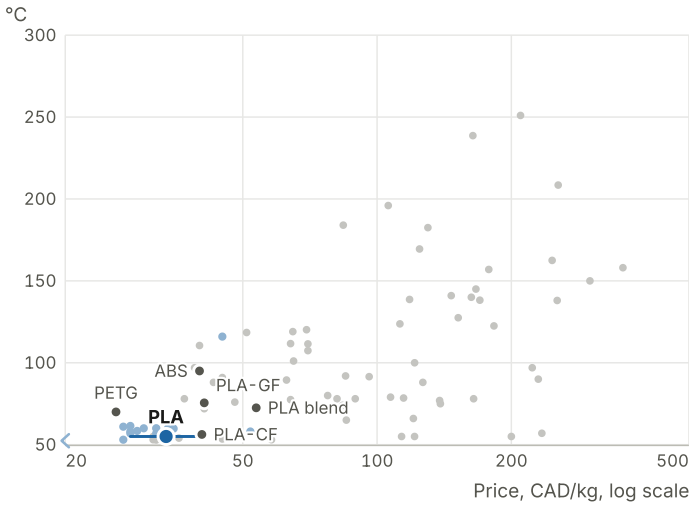
PLA: median, middle half as a cross · a PLA product · an alternative · another material

beyond the scale (its value on hover and in the table)

Tensile strength (XY) against tensile modulus (XY)



HDT at 0.45 MPa against price



Grey points are every other rigid material's medians, on frames that are the same on every rigid material's page; the alternatives are named, and a chevron at an edge marks a material beyond the frame.

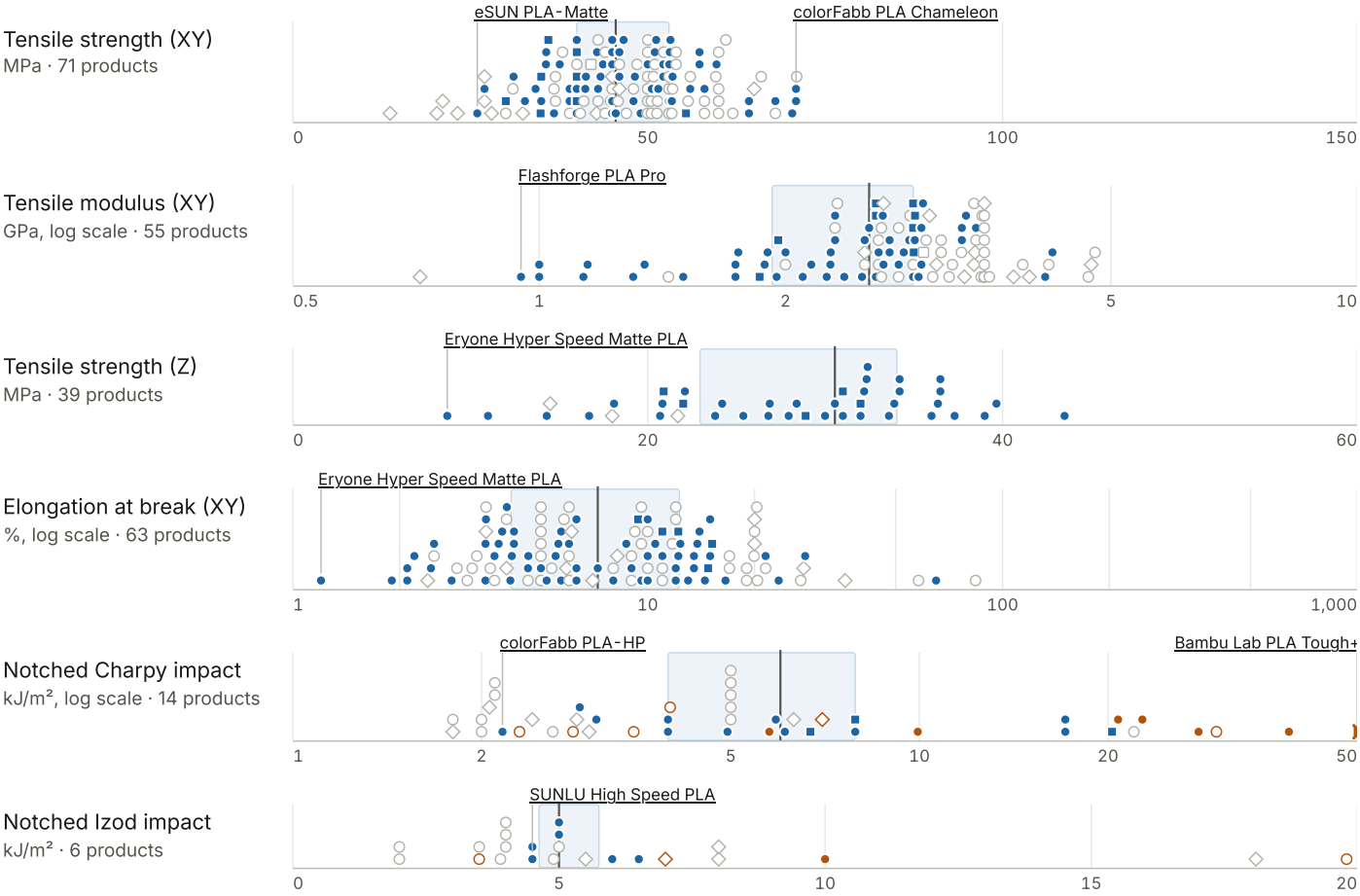
Mechanical properties

Printed flat (XY), PLA typically reaches a tensile strength of 45.5 MPa at a modulus of 2.53 GPa, and breaks at 7.23 % elongation. In bending, its flexural strength is typically 76 MPa at a flexural modulus of 2.87 GPa.

PLA's products spread most in elongation at break: the middle half runs 4.12–12 %

How strong, stiff and tough are its products, and which stand out?

- a product
- as published: orientation or load not stated, kept out of the median
- measured after annealing
- added filler or additive, kept out of the median
- sold as toughened (maker's statement)
- middle half and median
- beyond the scale (its value on hover and in the table)



One mark per product, on the scale the property has on every page (zoomed where its products sit close together, as its label says); the band is the middle half of the products ranked and the line their median.

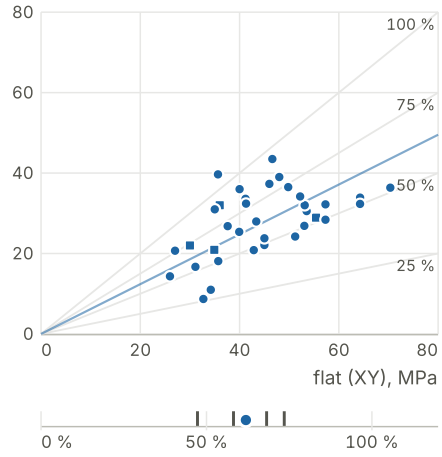
Across the layers (Z), PLA keeps 62 % of its flat strength and 89 % of its flat stiffness

What does a printed part lose across the layers (Z) against flat (XY), and is that usual?

a product · measured after annealing · equal share: 100, 75, 50 and 25 % · PLA's median share · an alternative

Tensile strength

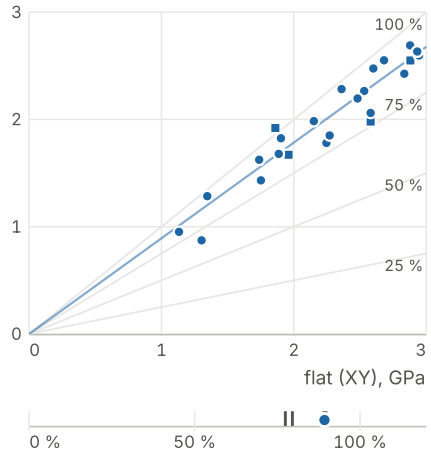
Z keeps 62 % of XY, median of 38 products



Share kept: ABS 73 % · PETG 68 % · **PLA 62 %** · PLA-CF 58 % · PLA-GF 47 %

Tensile modulus

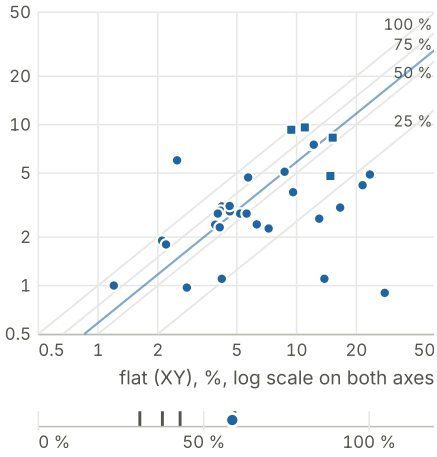
Z keeps 89 % of XY, median of 28 products



Share kept: PETG 90 % · **PLA 89 %** · ABS 89 % · PLA-GF 80 % · PLA-CF 77 %

Elongation at break

Z keeps 59 % of XY, median of 35 products



Share kept: PETG 59 % · **PLA 59 %** · PLA-CF 43 % · ABS 38 % · PLA-GF 31 %

Each point is a product that publishes both from one sheet: below the 100 % ray it keeps less across the layers than flat. Under each plot, the median share its alternatives' products keep.

Flat and across the layers

Property	Printed flat (XY)		Across the layers (Z)		Z as a share of XY
	Median	Products	Median	Products	
Tensile strength MPa	**45.5 MPa** 40 MPa–53 MPa	71	**30.6 MPa** 22.9 MPa–34 MPa	39	**62 %** 38 products
Tensile modulus GPa	**2.53 GPa** 1.93 GPa–2.87 GPa	55	**1.98 GPa** 1.67 GPa–2.48 GPa	29	**89 %** 28 products
Elongation at break %	**7.23 %** 4.12 %–12 %	63	**2.89 %** 2.28 %–4.85 %	35	**59 %** 35 products
Flexural strength MPa	**76 MPa** 69.5 MPa–84 MPa	63	**54 MPa** 42 MPa–59 MPa	17	**66 %** 16 products
Flexural modulus GPa	**2.87 GPa** 2.43 GPa–3.1 GPa	59	**2.41 GPa** 2.07 GPa–2.74 GPa	17	**88 %** 17 products

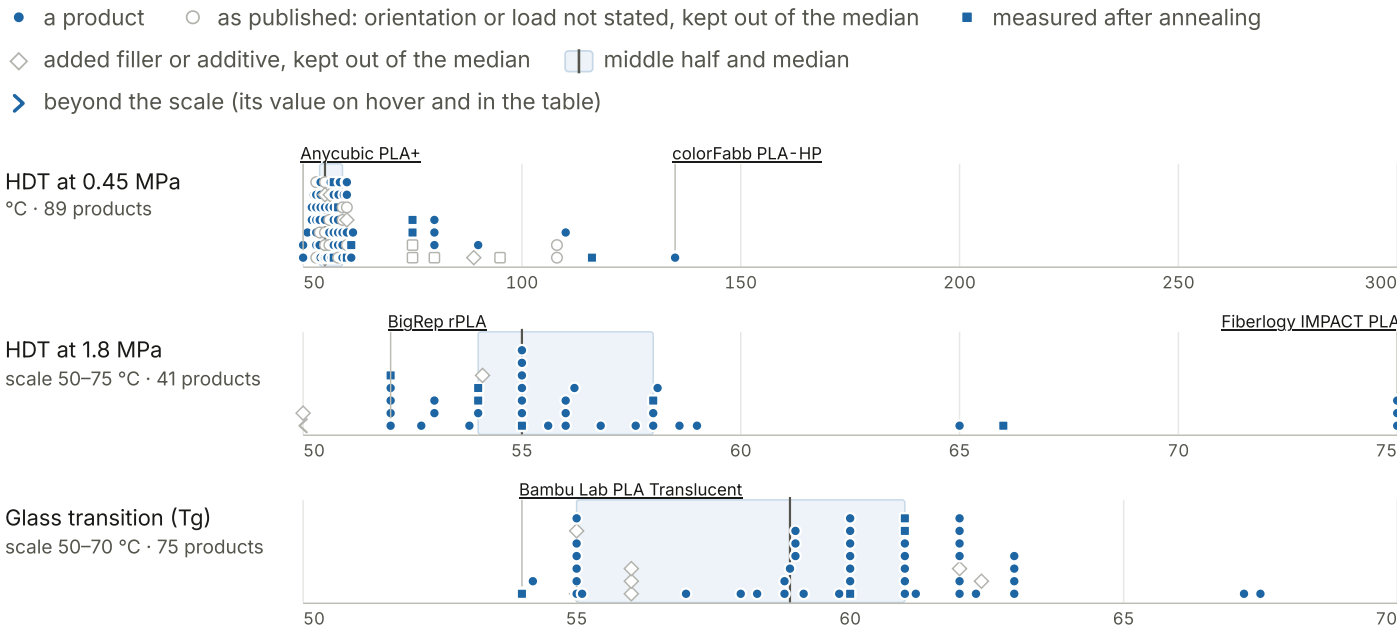
The middle half of the products under each median. The share is each product's Z value as a part of its XY value, from one sheet and one state, and the median of those shares. How XY and Z differ.

Thermal properties

PLA is amorphous as printed, so it softens at its glass transition, typically 59 °C: its HDT is 55 °C under 0.45 MPa and 55 °C under 1.8 MPa. Nine products publish 75 to 135 °C at 0.45 MPa, three of them after annealing. The highest are colorFabb PLA-HP (135 °C), Spectrum PLA Matt (116 °C, annealed at 90 °C for 4 h) and FormFutura Volcano PLA 150C (110 °C). As published, Vicat softening is typically 60 °C and melting typically 163 °C.

Most PLA products deflect at 54–59 °C under 0.45 MPa

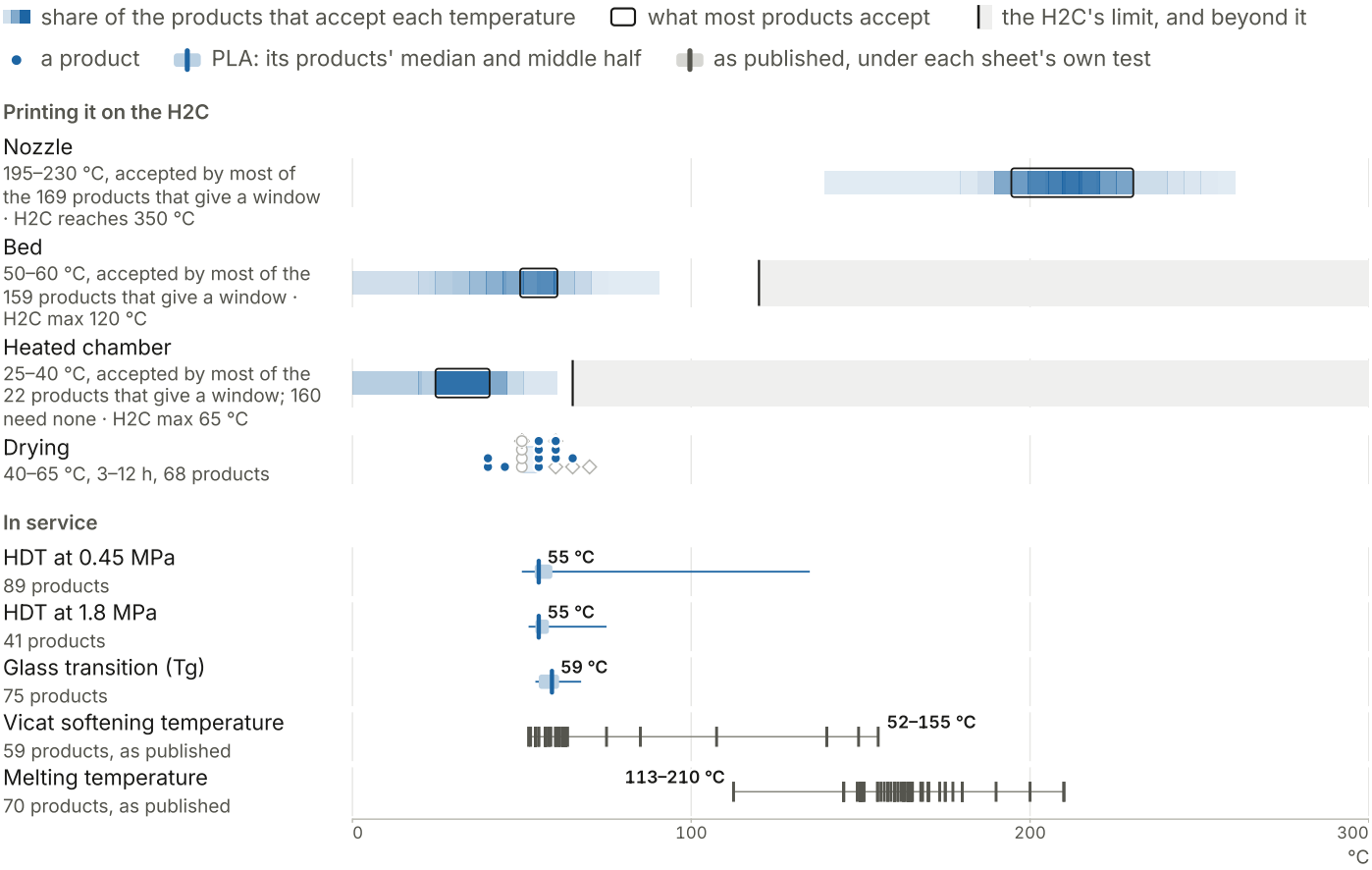
Where do its products soften under load?



One mark per product, on the scale the property has on every page (zoomed where its products sit close together, as its label says); the band is the middle half of the products ranked and the line their median.

PLA softens near 55 °C, below the H2C's 65 °C chamber

What does printing it on the H2C take, and how hot can a part get?



One temperature axis for the settings a print needs and the temperatures a part meets. A setting's shade is how many of the products whose sheets give a window accept each degree; the H2C's limit closes its row.

Setting	Most PLA products	Bambu Lab's Filament Guide for PLA
Nozzle	195–230 °C most products that give one	190 - 230 °C
Bed	50–60 °C most products that give one	Cool Plate (35 - 55 °C) Smooth PEI Plate (55 - 65 °C) Textured PEI Plate (55 - 65 °C) Cool Plate SuperTack (30-45 °C)
Heated chamber	none; sheets that give one say 25–40 °C 160 of 195 products	enclosure not needed
Drying	40–65 °C, 3–12 h most advise it only for wet filament	optional: 50 °C for 8 h
Nozzle type	standard for most	any

Setting	Most PLA products	Bambu Lab's Filament Guide for PLA
	6 of 195 products' sheets ask for one; 34 say a standard nozzle is enough; Bambu Lab's guide says any for 140 more; the rest are silent	
AMS	Bambu Lab PLA Tough+: "AMS, AMS 2PRO, AMS HT, AMS Lite"; the other 194: not stated	—

Where a product's own sheet is silent, its setting comes from a sheet it shares or from Bambu Lab's guide; each product's card says which.

Hardened nozzle. Six products need one: eSUN eStars-PLA, Nobufil PLAx, Protopasta Stainless Steel PLA, colorFabb steelFill, Protopasta Magnetic Iron PLA and Raise3D Hyper Speed PLA Pro.

After printing

15 products publish values measured after annealing, on schedules from 50 °C for 8 h to 110 °C for 15 min.

> "Measure dimensions of the part prior to annealing and again after, to determine shrink."
> MatterHackers Tough PLA, data sheet, p. 1

> "Typical time to anneal parts with wall -thickness of 0.125 in (~3.18 mm) is around 20 minutes, but this time is dependent on wall thickness."
> 3D-Fuel Pro PLA, data sheet, p. 1

> "As the name suggests, Volcano PLA 150C features an outstanding long term heat resistance of 150 °C, achieved without annealing or post-processing, for non-load-bearing parts."
> FormFutura Volcano PLA 150C, data sheet, p. 1

> Wet sanding reduces heat damage. Cyanoacrylate bonding is feasible; routine solvent smoothing is unsuitable.
> after Prusa Research, PLA | Prusa Knowledge Base

What makers say about printing it

Warping and shrinkage 22 makers

> "Fast printing up to 300 mm/s, low warp and a consistent surface finish."
> Polymaker HT PLA Pro, data sheet, p. 1

Odour and emissions 16 makers

> "Odorless during printing and fully biodegradable, making it safe for home use and the environment."
> eSUN PLA-Lite, maker's product page or guide, p. 1

Printing advice 19 makers

> "Remote printing needs to reduce the printing speed (≤40mm/s) to prevent potential feeding issue"
> eSUN ePLA ST, data sheet, p. 2

On PLA in general

> Low printing difficulty; limited warping and good small-feature reproduction. Difficulty
> after Prusa Research, PLA | Prusa Knowledge Base

Products

195 products from 40 makers; the most from eSUN (21), Spectrum (19) and SUNLU (14).

- Most documented: Bambu Lab PLA Basic, and 5 others that publish as much.
- Highest tensile strength (XY): colorFabb PLA Chameleon, 70.9 MPa (colorFabb PLA Vertigo prints the same sheet).
- Highest tensile strength (Z): Polymaker PolyLite PLA, 43.5 MPa, and it keeps the most strength across the layers, 93 % of its flat value.
- Highest HDT at 0.45 MPa: colorFabb PLA-HP, 135 °C.
- Highest notched Charpy impact: Bambu Lab PLA Tough+, 72.3 kJ/m², sold as toughened and measured after annealing at 50 °C for 8 h.
- Lowest price: eSUN PLA HF, 18 CAD/kg.

These are published values, not recommendations: the selector judges each product against your requirements.

Uses, limitations and chemicals

What it is sold for

> "Perfect for both large-scale additive manufacturing and small-batch prototyping."
>
> FormFutura Bulk PLA, data sheet, p. 1

> "Polymaker™ HT-PLA Pro is a tough PLA engineered for functional parts that need to survive real-world use."
>
> Polymaker HT PLA Pro, data sheet, p. 1

> "It combines the ease of printing that PLA is known for with significantly improved impact resistance, ductility and overall toughness, making it ideal for functional prototypes, mechanical components, snap-fits, brackets and end-use parts subject to dynamic or shock loads."
>
> Extrudr PLA TOUGH, data sheet, p. 1

> Detailed display models and early prototypes; low-load indoor parts.
>
> after Prusa Research, PLA | Prusa Knowledge Base

What makers warn about

> "The process may be less consistent on smaller nozzles (<. 4mm) and/or Bowden Type machines"
>
> Protopasta Stainless Steel PLA, data sheet, p. 1

> "For an outdoor usage, we recommend that you paint your print or submit them to an anti UV treatment, such as our Nanovia smoothing solution."
>
> Nanovia PLA EF 3D850, data sheet, p. 1

> "The combination of being both low UV resistance and having low-temperature resistance means that it's not ideal for outdoor use."
>
> Prusa Research Prusament PLA by Prusa Polymers, maker's product page or guide, p. 1

> "The material is not easy to bend as the strong rigidity."
>
> eSUN eTwinkling, data sheet, p. 2

> “Please note: each production run will entail slight deviation in color due to the deviation in the filament that is used for recycling.”
> colorFabb PLA Regrind, data sheet, p. 1

> Repeated impact, outdoor exposure or elevated service temperature governs.
> after Prusa Research, PLA | Prusa Knowledge Base

Chemicals and environment

Exposure	What makers say	Statements
Acid resistance	mostly not resistant: 25 not resistant, 7 resistant, 2 without a verdict, 1 limited	35
Alkali resistance	mostly not resistant: 7 not resistant, 2 without a verdict	9
Organic solvent resistance	mostly limited: 5 limited, 1 no data	6
Oil and grease resistance	mostly limited: 5 limited, 1 no data, 1 without a verdict	7
Water solubility	insoluble	5
Flammability	mostly flammable: 4 flammable, 1 without a verdict	5
UV and outdoor	stated without a verdict	6
Food contact claims	stated without a verdict	1

Of PLA in general:

> UV exposure and heat can degrade performance; unmodified PLA is not a weathering-grade default. UV and outdoor
> after Prusa Research, PLA | Prusa Knowledge Base

> Industrial compostability does not imply breakdown in home compost.
> Disposal and circularity
> after Prusa Research, PLA | Prusa Knowledge Base

> Polymer-level food-contact statements do not certify a printed article or its cleanability.
> Food contact claims
> after Prusa Research, PLA | Prusa Knowledge Base

> Material extrusion can release particles and volatile compounds. Assess ventilation and source capture; low odour does not establish low exposure. Protect against hot surfaces and machining dust. Safety
> after CDC / NIOSH, Potential Hazards of Additive Manufacturing

> Original study examines mineral-oil exposure coupled with low-cycle loading. The exposure and loading protocol must be matched; a generic oil-resistance label is insufficient.
> Oil and grease resistance
> after Głowacki et al., Polymers, Change in the Low-Cycle Performance on the 3D-Printed Materials ABS, ASA, HIPS, and PLA Exposed to Mineral Oil

Makers rarely give the concentration, temperature or duration, so these are indications, not design data. Each product's statements are on its card.

Reference data for plain PLA resin, where no maker of these products speaks: moisture sensitivity, limited (moisture content before melt processing, absorbs). Fillers, pigments and printing change it, so it never stands for these products.

Density and price

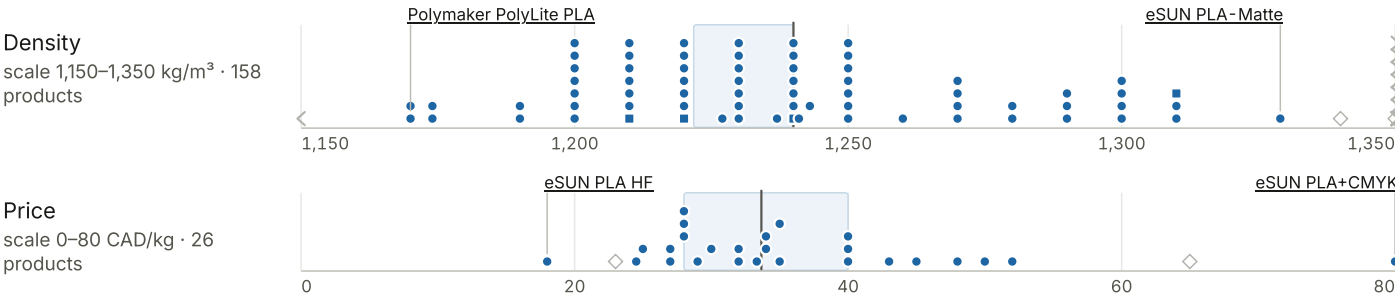
PLA's density is typically 1,240 kg/m³, the middle half 1,220–1,240 kg/m³, across 158 products. PLA typically costs 34 CAD/kg, the middle half 28–40 CAD/kg, across the 26 products with a regular price in the sample; the lowest is eSUN PLA HF at 18 CAD/kg.

Most PLA costs 28–40 CAD/kg

How dense is it, and what does it cost?

● a product ■ measured after annealing ◇ added filler or additive, kept out of the median ▯ middle half and median

› beyond the scale (its value on hover and in the table)



One mark per product, on the scale the property has on every page (zoomed where its products sit close together, as its label says); the band is the middle half of the products ranked and the line their median.

Regular prices, never sale prices, sampled from 9 Canadian shops and 7 foreign shops between 10 Sep and 7 Oct 2026; shipping and tax are not included.

Other properties, as published

Each value is under its own sheet's test conditions, so these are listed, not ranked. Each product's values, with their conditions and documents, are on its card.

Property	Products	Lowest–highest
Melt mass-flow rate	107	2.13–70 g/10 min
Water absorption	37	0.13–1.8 %
Elongation at yield	22	0.8–13 %
Mould shrinkage	18	0.1–0.67 %
Decomposition temperature	16	250–375 °C
Hardness	15	80–86.5 Shore (scale not specified by source) 72–87 Shore D
Impact strength, test unclear	13	2.1–17.6 kJ/m²
Crystallization temperature	12	77.2–130 °C
Tensile strain at strength	7	2.07–15 %
Continuous service temperature	6	45–50 °C

Property	Products	Lowest–highest
Interlayer adhesion strength	4	17–41 MPa
Melt volume-flow rate	4	8–27 cm³/10 min
Moisture content	4	0.1–0.26 %
Flexural elongation at break	3	3.3–10 %
Flexural stress at conventional deflection	2	79–80.1 MPa

Sources

281 documents from 41 publishers (240 data sheets, 35 makers' product pages and guides, 2 safety data sheets, 1 printer document, 1 reference, 1 safety guidance document and 1 peer-reviewed study), read 10 Sep–5 Oct 2026. Each value on PLA's Products page names its document and page, and links to the original.

Publisher	Documents	Values	Products
Spectrum	30	146	22
eSUN	25	202	21
Polymaker	22	371	13
Fiberlogy	21	116	9
colorFabb	20	168	12
SUNLU	18	223	14
FormFutura	13	95	13
Bambu Lab	11	185	6
Prusa Research	11	111	5
Extrudr	11	91	6
Eryone	8	99	8
MatterHackers	7	33	4
and 29 more publishers, with 82 documents			

Bambu Lab, H2C Filamant Printing Guide: print settings for 191 products.

Bambu Lab's Filament Guide: print settings for 173 products whose own sheets are silent.

Data-sheet values describe test specimens, not design allowables. Confirm the exact product, its conditioning and your part's geometry with your own print.

Filament Atlas, release `7f3f14384dab`. Documents read 2026-09-10 to 2026-10-05; built `2026-10-17`.

Summary by Claude Opus (agent), 2026-10-17. Draft, not yet reviewed by the owner.

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